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# LEGAL MATE AI: A DOMAIN SPECIFIC INDIAN LEGAL AI MODEL FOR LEGAL GUIDANCE AND COMPLAINT ASSISTANCE

**MINI PROJECT REPORT(21PBAI53)**

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# BONAFIDE CERTIFICATE

Certified that this project report "**LEGAL MATE AI: A DOMAIN SPECIFIC INDIAN LEGAL AI MODEL FOR LEGAL GUIDANCE AND COMPLAINT ASSISTANCE"**, is the Bonafide work of **R.VARSHAN - 24619133**, **P.PRAVEEN KUMAR - 24619125**, **P.NAVEEN KUMAR - 24619124** carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported here does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate

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# ABBREVIATIONS

| ABBREVIATION | FULL FORM |
|---|---|
| AI | Artificial Intelligence |
| NLP | Natural Language Processing |
| RAG | Retrieval-Augmented Generation |
| LLM | Large Language Model |
| OCR | Optical Character Recognition |
| API | Application Programming Interface |
| PDF | Portable Document Format |
| UI | User Interface |
| DB | Database |
| JS | JavaScript |
| CSS | Cascading Style Sheets |

# ABSTRACT

Legal Mate AI is an AI-based legal assistance and companion system designed to provide accessible, source-grounded preliminary legal information based on Indian laws. Many citizens face difficulties understanding legal terminology, identifying applicable laws, knowing their rights, and following complaint procedures. The proposed system combines Artificial Intelligence, Natural Language Processing (NLP), and Retrieval-Augmented Generation (RAG) to understand users' legal queries, identify the relevant legal domain, and retrieve appropriate information from a curated Indian legal knowledge base. The project also focuses on developing a domain-specific Indian Legal AI model trained using curated legal problem descriptions, categories, intents, and question-answer examples. The system can explain legal rights, identify potentially relevant Acts and sections, analyze uploaded legal documents, generate structured complaint drafts, provide evidence and document checklists, and guide users through possible next steps. English and Tamil support is proposed. The system provides legal information and procedural guidance, not professional legal representation.

**Keywords :** *Artificial Intelligence, Natural Language Processing (NLP),Retrieval-Augmented Generation (RAG),Custom Legal AI Model,Legal Information Retrieval, Complaint Generation*

# CHAPTER 1

# INTRODUCTION

## 1.1 OVERVIEW OF THE PROJECT

Legal Mate AI is an intelligent legal-information assistance platform developed to help users understand legal documents, legal terminology, and important clauses in a simple and organized manner. Legal documents such as agreements, contracts, notices, petitions, affidavits, rental agreements, employment documents, and other official records often contain complex legal language. Ordinary users may find it difficult to identify important information, understand their rights and responsibilities, or locate clauses related to a particular question. Legal Mate AI reduces this difficulty by combining document processing, artificial intelligence, semantic search, and natural-language interaction in a single application. The main purpose of the project is to make legal information more accessible to students, citizens, and users without a legal background. Instead of reading an entire document manually, users can upload a legal document and ask questions about its contents. The system processes the uploaded document, extracts available text, organizes the content into smaller meaningful sections, and retrieves the parts most relevant to the user's question. It then provides a clear response based on the information available in the uploaded document. Legal Mate AI follows a document-grounded approach, meaning that it does not rely only on general knowledge while answering a question. It first searches the uploaded document for relevant passages and uses those passages as context for generating the answer. This keeps the response connected to the actual document provided by the user. For example, if a user uploads a rental agreement and asks about the notice period, the system can retrieve the clause related to termination or notice and explain it in simple language. Similarly, a user can upload an employment agreement and ask about salary terms, working hours, leave policy, confidentiality clauses, or termination conditions. The system begins with a user interface that allows users to sign in and access the main dashboard. From the dashboard, users can upload legal documents in supported formats. The system extracts readable text from the uploaded document and separates the content into smaller passages so that important sections can be searched efficiently. After processing the document, Legal Mate AI creates an indexed representation of the text, allowing the system to perform semantic search and identify relevant information based on meaning rather than only exact keyword matching. When a user asks a question, the system compares the question with indexed document passages and retrieves the most relevant content. The retrieved passages are then provided to the AI response component, which generates an understandable explanation, summary, or answer based on the available document context. Another important feature of Legal Mate AI is document summarization. Legal documents can be lengthy and difficult to read completely, so the system can provide a concise overview of the document, helping the user understand its purpose, major sections, obligations, conditions, and important terms. The application also provides a legal query interface where users can ask focused questions in natural language, such as “What is the notice period?”, “What are the responsibilities of the tenant?”, “Does this agreement mention a penalty?”, or “What happens if either party terminates the contract?”. The system retrieves relevant information from the uploaded document and presents it in a simple format. Where possible, the response can include the

related source passage or section reference so that the user can verify the result in the original document. Legal Mate AI can also include a document and query history feature, allowing users to access previously uploaded documents, summaries, and questions. This feature helps users organize legal-document research and revisit earlier queries without repeating the process. Since legal documents can contain sensitive information, the system should use suitable authentication, controlled access, secure storage, and careful handling of uploaded data. Legal Mate AI is designed as a legal-information support system and not as a replacement for a lawyer, advocate, or legal professional. The responses generated by the system are intended to help users understand document content and identify relevant clauses; they are not legal advice. Legal matters may depend on facts, jurisdiction, current laws, court decisions, and professional interpretation. Therefore, users should consult a qualified legal professional before making legal decisions, signing important documents, filing a case, or taking action based on the system's output. Overall, Legal Mate AI combines artificial intelligence, document text extraction, semantic retrieval, question answering, summarization, source reference, and document history into one platform. The project aims to make legal documents easier to understand, easier to search, and more accessible to users.

### 1.2 PROBLEM STATEMENT

Legal documents such as agreements, contracts, notices, affidavits, petitions, and employment documents often contain complex legal terminology and lengthy clauses. Many students and ordinary users find it difficult to understand the meaning of these documents, identify important conditions, and locate information related to their rights and responsibilities. Manual reading

of a long legal document requires time and basic legal knowledge. Existing online search systems may provide general legal information, but they may not explain the content of a specific document uploaded by a user.

## The major problems are:

1. **Complex legal language:** Legal documents contain formal terms, technical phrases, and complex sentence structures that are difficult for non-legal users to understand.
2. **Time-consuming document review:** Users must read several pages manually to find important clauses such as payment terms, notice periods, obligations, penalties, confidentiality, and termination conditions.
3. **Limited document-based assistance:** General AI chatbots may give broad answers but may not connect the response to the user's uploaded legal document.
4. **Difficulty locating relevant clauses:** Keyword-based searches may fail when the user does not know the exact legal terms used in the document.
5. **Lack of simple summaries:** Users may not receive a short and understandable overview of a long legal document.

6. **Risk of misunderstanding:** Incorrect interpretation of a legal clause can lead to confusion, missed deadlines, or poor decision-making.
7. **Privacy concerns:** Legal documents may contain confidential personal, financial, or contractual information that should be handled securely.

Therefore, there is a need for an intelligent legal-information platform that can accept legal documents, extract and organize their text, identify relevant passages, generate simple summaries, and answer user questions based on the uploaded document. Legal Mate AI is proposed to address this need by providing document-grounded legal assistance. The system helps users understand and search legal content more easily, while clearly informing them that the generated output is for general information only and does not replace advice from a qualified legal professional.

## 1.3 OBJECTIVES

The main objective of the project **"LEGAL MATE AI: A DOMAIN SPECIFIC INDIAN LEGAL AI MODEL FOR LEGAL GUIDANCE AND COMPLAINT ASSISTANCE"** is to develop an AI-powered platform that helps Indian citizens understand basic legal information, obtain guidance on common legal issues, and prepare structured complaint content in a simple and accessible manner.

- To develop a domain-specific AI model focused on Indian legal information and complaint assistance.
- To provide users with simple explanations of common legal concepts, rights, responsibilities, and procedures in India.
- To help users identify the general legal category related to their issue, such as consumer complaints, cybercrime, domestic issues, employment issues, property disputes, or public grievances.
- To assist users in preparing structured complaint drafts based on the information provided by them.
- To guide users about the basic details and documents that may be required while submitting a complaint.
- To provide step-by-step information about possible complaint channels, such as police stations, cybercrime portals, consumer forums, government grievance portals, or other relevant authorities.
- To allow users to upload and analyze legal documents or complaint-related documents for easier understanding.
- To extract and summarize important information from uploaded documents and present it in simple language.

- To provide document-based answers by retrieving relevant content from uploaded files.
- To maintain user query and document history for future reference, subject to privacy and access-control measures.
- To improve legal awareness among users who may not have access to immediate legal assistance.
- To ensure that the system clearly states that it provides general legal guidance and complaint assistance only, and does not replace advice from a qualified advocate or legal professional.

## 1.4 SCOPE OF THE PROJECT

The scope of the project is to develop an AI-powered legal assistance system that helps users understand their legal rights and access relevant Indian laws. The system uses Natural Language Processing (NLP), Machine Learning, and Retrieval-Augmented Generation (RAG) to identify legal issues and provide relevant legal information. It also supports English and Tamil responses, generates structured complaint drafts, and suggests necessary documents and possible next steps. The system aims to make legal information more accessible, simplify complaint preparation, and provide reliable preliminary legal guidance to users.

## 1.5 APPLICATIONS AND EXPECTED OUTCOMES

### Applications

- Provides basic legal guidance to users based on Indian laws.
- Helps users identify legal issues through AI-based query classification.
- Retrieves relevant legal acts, sections, and provisions using RAG technology.
- Generates structured complaint drafts for different legal issues.
- Supports English and Tamil to improve accessibility.
- Suggests supporting documents and possible steps for filing complaints.

### Expected Outcomes

- Improved accessibility to legal information for the general public.
- Faster identification of relevant laws and legal provisions.
- Simplified complaint preparation with structured drafts.
- Better understanding of legal rights and available remedies.
- Reduced time and effort required to find relevant legal information.

# CHAPTER 2

# LITERATURE REVIEW

## 2.1 LITERATURE OVERVIEW

The development of Artificial Intelligence, Natural Language Processing, machine learning, and information-retrieval systems has created new opportunities for improving access to legal information. Legal documents, laws, regulations, court decisions, and complaint procedures are often written in formal language that can be difficult for ordinary citizens to understand. In India, people may face challenges in identifying the correct authority, understanding the steps required to submit a complaint, preparing a complaint letter, and finding basic information about their legal rights. AI-based legal-assistance systems can support users by converting complex legal information into simpler explanations, identifying relevant legal categories, and guiding users through structured complaint preparation.

Existing legal-information platforms generally provide legal articles, case-law databases, legal search tools, and government portals for complaint registration. However, these resources may require users to know the correct legal terms, sections, procedures, and authorities in advance. General-purpose chatbots can provide broad answers to legal questions, but they may not always be based on Indian legal context, current legal provisions, or the user's specific complaint details. In addition, a general response may not guide a user in preparing a structured complaint or identifying the appropriate complaint channel.

Recent research in Retrieval-Augmented Generation has shown that combining document retrieval with language models can improve the relevance of generated answers. In this method, the system first retrieves relevant legal information from a trusted knowledge base or user-uploaded document and then uses that information to generate a response. This approach is useful for legal applications because it can connect the generated answer with relevant legal content instead of relying only on general language-model knowledge. It can also improve transparency by allowing users to review the source information related to the answer.

Natural Language Processing can also support legal-document summarization, legal-text classification, semantic search, question answering, and complaint drafting. Legal-text classification can help identify whether a user's issue is related to consumer protection, cybercrime, employment, property, domestic matters, public grievances, or another legal category. Semantic search can identify relevant information based on the meaning of a user's question instead of relying only on exact keywords. Summarization can help users understand long legal documents, notices, agreements, or complaint-related materials in a shorter and simpler form.

Legal Mate AI is designed by combining these concepts into a domain-specific Indian legal-assistance platform. The system aims to provide basic legal guidance, support complaint drafting, explain uploaded legal documents, and guide users toward possible complaint channels. The platform is intended to improve legal awareness and accessibility for users who may not have immediate access to legal assistance. However, the system must clearly state that

it provides general legal-information support only and does not replace the services, interpretation, or advice of a qualified advocate or legal professional.

## 2.2 LITERATURE SURVEY

### 1. Legal NLP: A Comprehensive Review (2023)

The study on Legal Natural Language Processing provides an overview of how Artificial Intelligence, Machine Learning, Deep Learning, and Transformer models can be applied to legal documents. It explains that legal documents are complex because they contain formal terminology, long sentences, references to Acts and sections, and context-dependent meanings. The study focuses on tasks such as legal-document classification, legal information retrieval, legal question answering, text summarization, and legal entity extraction. This work is relevant to Legal Mate AI because the proposed system uses Natural Language Processing to understand user queries, identify legal domains, retrieve relevant legal information, and provide preliminary legal guidance. A limitation of this approach is that the availability of high-quality legal datasets and domain-specific legal knowledge remains a challenge.

### 2. Named Entity Recognition Methods and Instruments Based on Improved Few-Shot Learning (2024)

This study focuses on Named Entity Recognition, a Natural Language Processing technique used to identify important entities from text. In legal documents, named entities may include names of persons, organizations, locations, dates, legal sections, case numbers, authorities, and government departments. The study uses approaches such as BERT, Few-Shot learning, and data augmentation to improve entity recognition when only a limited amount of labelled data is available. This is useful for Legal Mate AI because the system may need to identify important information from complaint descriptions and uploaded documents. For example, it can identify the complainant name, incident date, place of occurrence, involved party, and relevant authority. The limitation is that the study may be focused on a different language domain, so additional training and adaptation are required for Indian legal documents and regional-language support.

### 3. CHRExpert: An AI-Driven Court of Human Rights Expert Assistant (2025)

CHRExpert is an AI-based legal assistant designed to support legal research and analysis in the domain of human rights. The system uses advanced language models and transformer-based methods to understand legal questions, retrieve relevant legal information, and assist users in analysing complex legal content. The main objective of the system is to provide legal professionals and users with a more efficient way to access legal knowledge and identify relevant legal materials. This study is relevant to Legal Mate AI because it shows how an AI assistant can support legal guidance through question answering and information retrieval. However, the system is limited to a particular legal domain and jurisdiction. Legal Mate AI extends this concept by focusing on preliminary Indian legal guidance, complaint assistance, legal-domain identification, and structured complaint drafting.

**4. SmartLaw Advisor: AI-Powered Legal Consultation and Contract Analysis for Saudi Arabia (2026)**

SmartLaw Advisor is an AI-powered legal consultation and contract-analysis system developed for users who need assistance in understanding legal documents and contracts. The system uses large language models, retrieval-augmented generation, document processing, and vector search to retrieve relevant legal content and provide understandable answers. It can analyse contract clauses, explain legal terms, and support users in identifying important conditions in documents. This work is relevant to Legal Mate AI because both systems aim to make legal information more accessible to ordinary users. Legal Mate AI uses a similar concept of source-grounded retrieval but is designed for the Indian legal context. It focuses not only on legal-document analysis but also on complaint preparation, evidence and document checklists, possible next steps, and support in English and Tamil. A limitation of SmartLaw Advisor is that its knowledge base is specific to Saudi Arabian law; therefore, it cannot be directly applied to Indian laws and legal procedures.

## 2.3 RESEARCH GAP

Existing research in Legal Natural Language Processing (NLP), named entity recognition, AI-based legal assistance, and Retrieval-Augmented Generation (RAG) has demonstrated the potential of artificial intelligence in improving legal information access. However, several limitations remain in providing integrated legal assistance tailored to Indian laws and complaint procedures. Existing approaches primarily focus on individual tasks such as legal-text classification, entity extraction, legal question answering, or contract analysis [1]–[4]. Furthermore, jurisdiction-specific systems may not be directly applicable to the Indian legal framework, while general-purpose chatbots may produce unsupported information and lack structured complaint-generation capabilities. Limited multilingual support may also restrict accessibility for users who prefer regional languages. Therefore, Legal Mate AI proposes an integrated framework combining legal intent classification, RAG-based retrieval from an Indian legal knowledge base, English and Tamil responses, structured complaint drafting, document checklists, and guidance on possible next steps. The proposed system aims to improve the accessibility and usability of preliminary legal information for Indian citizens.

Additionally, the system aims to simplify the process of identifying relevant legal provisions and understanding possible remedies for common legal issues. By combining information retrieval with context-aware response generation, it seeks to provide more relevant and understandable guidance. The proposed framework also emphasizes user-friendly interaction, reliable source references, and responsible use of AI to support legal awareness and informed decision-making.

**Table 2.1 Research Gap Analysis**

| Ref. | Existing Approach | Identified Research Gap | Proposed Approach |
|---|---|---|---|
| [1] | Legal NLP techniques | Limited focus on an integrated citizen-oriented legal assistance workflow. | Combines legal query classification, retrieval, and complaint drafting. |
| [2] | Named entity recognition | Requires adaptation to Indian legal documents and complaint-specific information. | Extracts relevant complaint details to support structured drafting. |
| [3] | Human-rights legal assistant | Domain-specific coverage may not address broader Indian legal issues. | Supports multiple Indian legal categories within the available knowledge base. |
| [4] | Saudi Arabian legal consultation and contract analysis | Jurisdiction-specific knowledge may not apply to Indian laws and procedures. | Uses Indian legal documents and jurisdiction-specific information. |

Research Contribution: The proposed Legal Mate AI system addresses the identified gaps by integrating NLP-based legal intent classification, RAG-based legal information retrieval, multilingual response generation, and structured complaint assistance into a single platform designed for the Indian legal context. The system is intended to provide preliminary legal information and does not replace professional legal advice.

# CHAPTER 3
# SYSTEM DESIGN

## 3.1 EXISTING SYSTEM

Traditional legal research relies on manual reading, keyword search, and professional consultation. Search engines and public legal portals can return a large number of results, but they may not identify the clauses most relevant to a user's question. Generic AI chat tools can explain legal concepts, yet they may not be connected to the specific document the user wants to understand.

**Limitations of the Existing System**

1. Complex legal terminology: formal drafting and technical terms can be difficult for non-specialists to interpret.
2. Fragmented workflow: upload, search, summarization, and question answering are often handled by separate tools.
3. Limited traceability: generic responses may not show which document passages support the answer.
4. Manual review effort: users must repeatedly scan long documents to locate key clauses.
5. Risk of over-reliance: automated output can be misunderstood as legal advice if limitations are not clearly stated.

## 3.2 PROPOSED SYSTEM

Legal Mate AI is a domain-specific Indian legal guidance and complaint-assistance platform designed to help citizens understand common legal issues in a simple and accessible manner. The system accepts user queries written in natural language and analyses the issue using Natural Language Processing. It identifies the relevant legal domain, such as consumer complaints, cybercrime, employment matters, property issues, domestic concerns, or public grievances. After identifying the legal category, the system retrieves relevant information from a curated Indian legal knowledge base. This knowledge base may contain simplified legal information, complaint procedures, basic legal rights, possible authorities, relevant Acts and sections, required documents, and commonly asked questions. The system uses a Retrieval-Augmented Generation approach, where relevant information is retrieved before an AI response is generated. This helps provide source-grounded preliminary guidance instead of only general answers . Legal Mate AI can assist users in preparing structured complaint drafts by collecting important details such as names, dates, locations, incident descriptions, and available evidence. It can also suggest possible evidence or document checklists and guide users about possible next steps. The system supports English and Tamil to improve accessibility. Legal Mate AI provides general legal-information and procedural guidance only; it does not replace advice from a qualified advocate or legal professional.

## 3.3 SYSTEM ARCHITECTURE

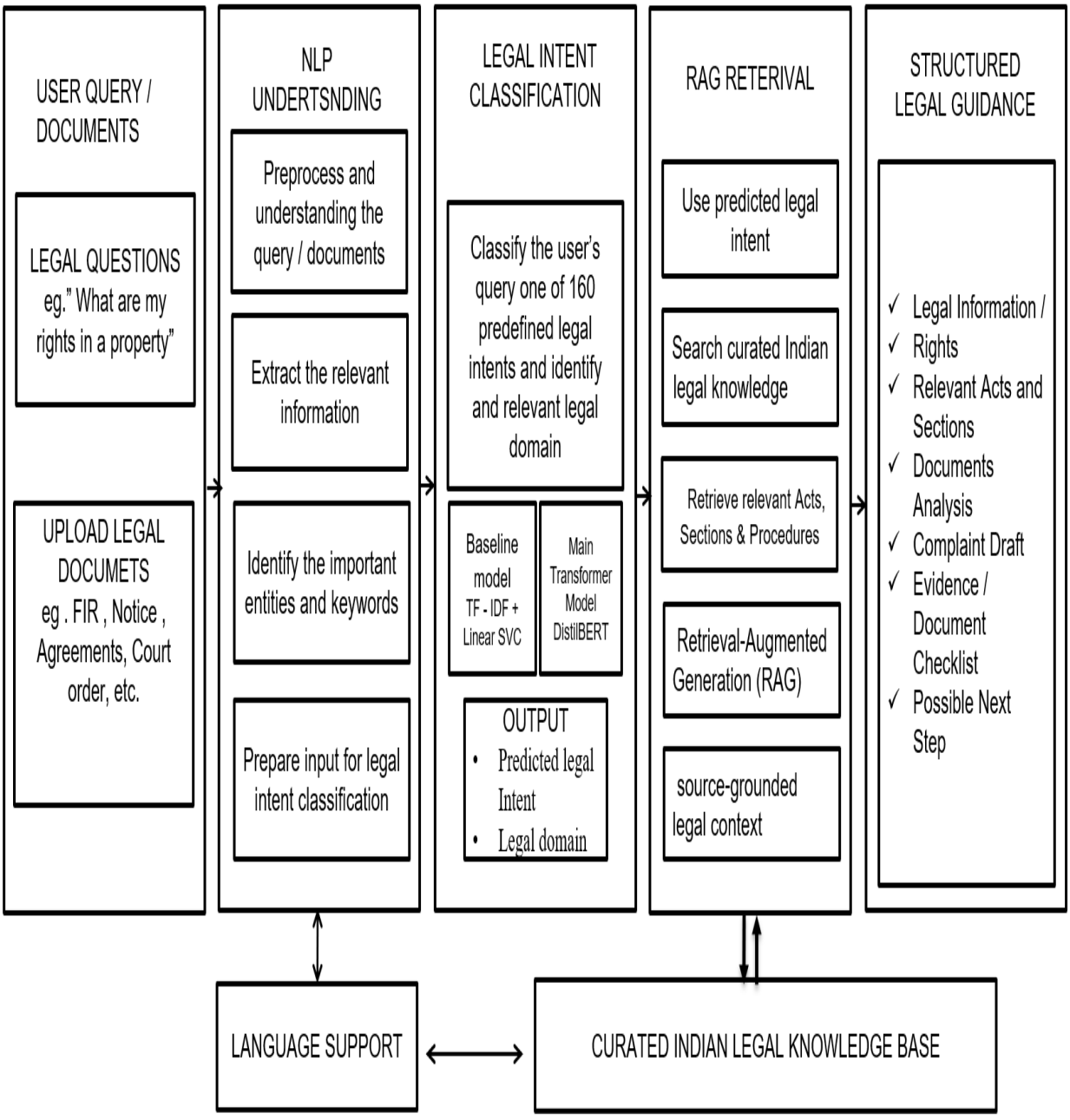



**Figure 3.1** Legal Mate AI system architecture

**Data Flow Sequence :**

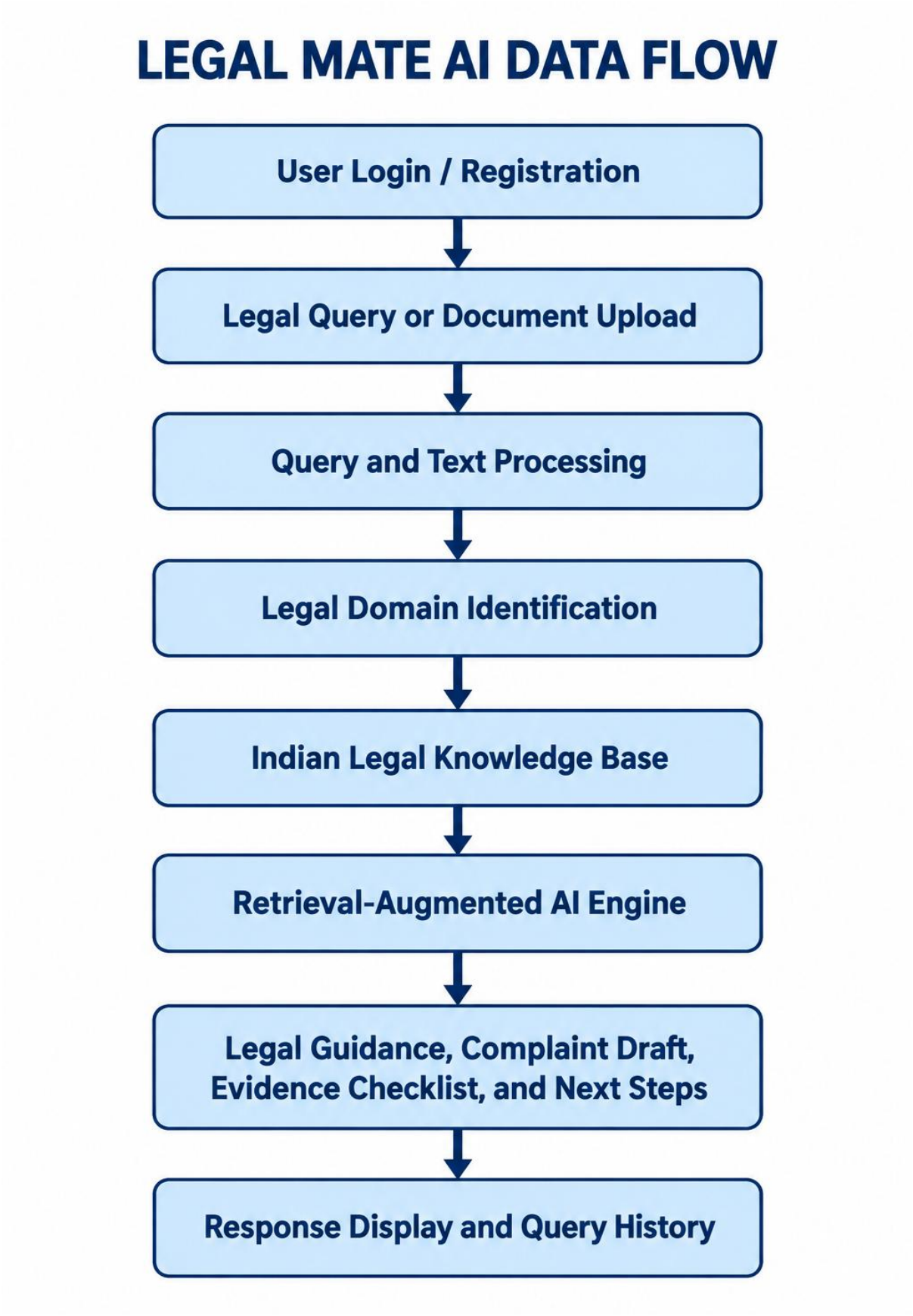



**Figure 3.2** Legal Mate AI data flow

Users submit a legal query or document. Legal Mate AI identifies the legal domain, retrieves relevant Indian legal information, and provides guidance, complaint support, evidence checklists, and next steps.

## 3.4 SOFTWARE REQUIREMENTS

**Table 3.4.1** Software Tool

| Software | Description |
|---|---|
| Operating System | Windows 10/11 or Ubuntu 20.04 or later |
| Programming Languages | Python, HTML, CSS and JavaScript |
| Development Environment | Visual Studio Code or another suitable code editor |

**Table 3.4.2** Libraries and Component

| Component | Specification / Purpose |
|---|---|
| Document Parser | Extracts text from supported legal documents |
| OCR Engine | Extracts text from scanned pages and images |
| Embedding and Retrieval | Indexes passages and retrieves relevant document context |
| Language Model | Generates source-grounded summaries and responses |
| Security Controls | Supports authentication, authorization, and protected document access |

**Table 3.4.3** Frontend

| Software | Description |
|---|---|
| Web Browser | Google Chrome, Firefox, or Microsoft Edge |
| Web Technologies | HTML, CSS, and JavaScript for the user interface |

**Table 3.4.4** Data Management

| Component | Description |
|---|---|
| Document Storage | Stores uploaded documents and extracted text according to configured access controls |
| History | Supports retrieval of prior documents, queries, and summaries |

# CHAPTER 4
# IMPLEMENTATION

## 4.1 METHODOLOGY

Legal Mate AI follows a structured methodology to provide preliminary Indian legal guidance and complaint assistance. The process begins when a user registers or logs in and submits a legal query or readable legal document. The user may describe consumer complaints, cybercrime, employment, property, domestic concerns, or public grievances. The system processes the input using Natural Language Processing techniques to identify important keywords, the nature of the issue, and the relevant legal domain. It then retrieves related information from a curated Indian legal knowledge base containing basic legal rights, complaint procedures, relevant Acts or sections, possible authorities, required documents, and frequently asked questions. Legal Mate AI uses Retrieval-Augmented Generation to produce source-grounded responses. It first retrieves relevant legal information and then generates a simple explanation based on the retrieved content. This approach improves relevance and reduces dependence on general responses. For complaint assistance, the system collects details such as names, dates, location, issue description, and available evidence. It generates a structured complaint draft, evidence checklist, and possible next steps. Finally, it displays the guidance with a disclaimer and stores the response in the user's query history. Legal Mate AI provides legal-information guidance only and does not replace professional legal advice.

### Dataset Description

The system is designed to operate on documents uploaded by the user rather than a dedicated training dataset. Publicly available legal material and sample documents may be used only for functional testing. No confidential legal document should be used for testing without appropriate authorization and data-protection controls.

**Step 1** - User sign-in: The user accesses the application through the login interface.

**Step 2** - Document upload: The user uploads a supported legal document.

**Step 3** - Text extraction: The application obtains readable text from the document or OCR output.

**Step 4** - Processing: The extracted text is cleaned, split into passages, and indexed.

**Step 5** - Query submission: The user asks a focused question about the uploaded content.

**Step 6** - Retrieval and response: Relevant passages are retrieved and used to generate a source-grounded answer.

**Step 7** - History: The document and query history can be reviewed by the user, subject to configured storage controls.

## 4.2 IMPLEMENTATION OVERVIEW

The implementation of Legal Mate AI focuses on a clear and responsible user experience for legal-document understanding. The user interface supports document upload, search, document summaries, and conversational questions. The processing layer extracts and organizes text, while the retrieval layer identifies passages relevant to the user's query. The AI layer converts those passages into an understandable response without presenting the output as professional legal advice.

The project can be implemented using a web frontend, an application server, a document-processing component, an embedding and retrieval component, and a database or managed storage service. Specific tools may be selected according to the final deployment requirements.

## 4.3 IMPLEMENTATION MODULE

### 4.3.1 Authentication and User Interface Module

This module provides controlled access to Legal Mate AI and presents the dashboard, document library, and question interface. It validates required user inputs and provides clear navigation between upload, analysis, and history features.

### 4.3.2 Legal Document Upload Module

This module accepts supported document formats and performs basic validation before processing. It informs the user about accepted files and avoids treating an upload as legal advice or a legal conclusion.

### 4.3.3 Legal Document Text Extraction Module

This module extracts text directly from machine-readable documents. OCR is used only when an upload legal document is a scanned PDF or an image and its text cannot be selected. The

extracted output is checked before it is sent to later stages.

This module extracts text directly from machine-readable documents. OCR is used only when an uploaded legal document is a scanned PDF or an image and its text cannot be selected. The extracted output is checked before it is sent to later stages.

### 4.3.4 Document Chunking and Indexing Module

This module divides long documents into meaningful passages, stores metadata such as page or section information where available, and creates representations that support semantic search.

### 4.3.5 Legal Query and Retrieval Module

This module receives the user's question and retrieves the most relevant indexed passages. Retrieval helps the system focus the answer on the uploaded material instead of relying only on general language-model knowledge.

### 4.3.6 AI Response and Summarization Module

This module creates concise explanations, summaries, and document-grounded answers. It is instructed to state uncertainty, avoid unsupported claims, and direct the user to a qualified legal professional for matters requiring legal advice.

### 4.3.7 Source Reference Module

This module presents relevant passages or source labels with the response so that users can verify how the answer relates to their document.

### 4.3.8 Document History Module

This module lets users revisit uploaded documents, prior summaries, and earlier questions. Access and retention should follow the configured privacy and security requirements.

### 4.3.9 Safety and Disclaimer Module

This module communicates that Legal Mate AI provides general legal-information assistance. It does not replace a lawyer, predict the outcome of a matter, or provide jurisdiction-specific legal advice.

## 4.4 EXPERIMENT ANALYSIS

Functional testing evaluates whether documents can be uploaded, text can be extracted, questions retrieve relevant passages, and summaries remain understandable. The evaluation should also check response relevance, source linkage, handling of incomplete documents, and the visibility of the legal-advice disclaimer.

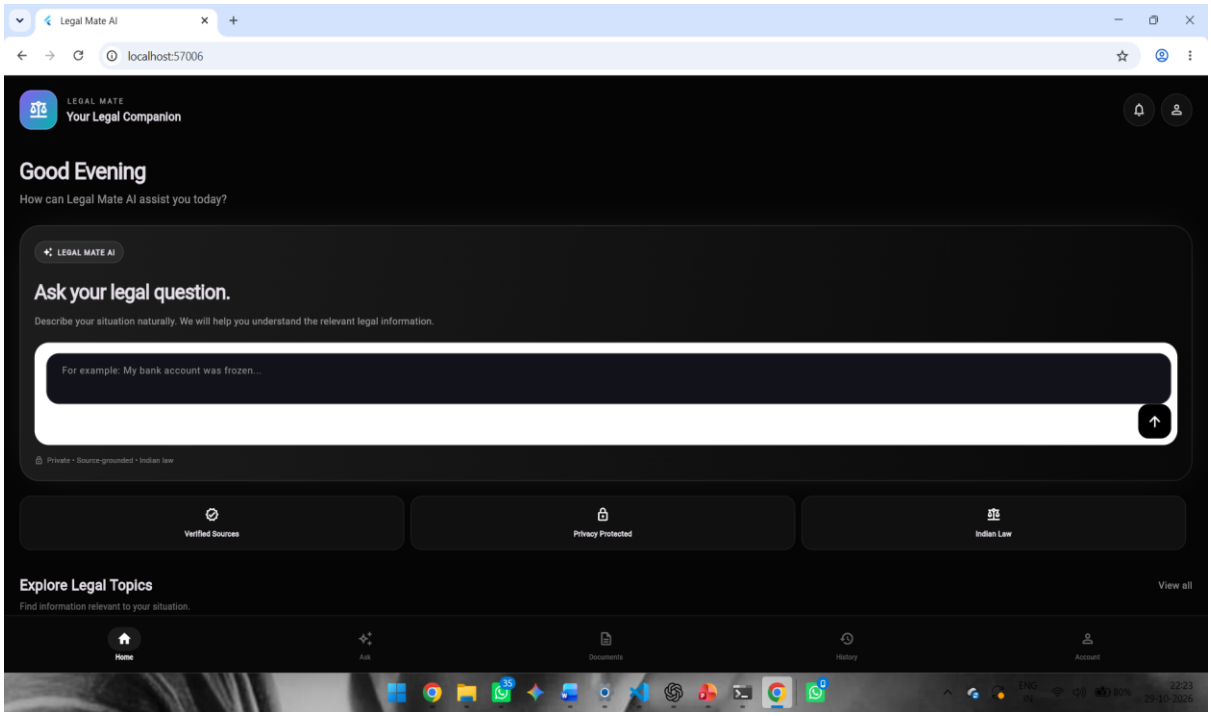



Figure 4.1 Legal Mate AI home Screen

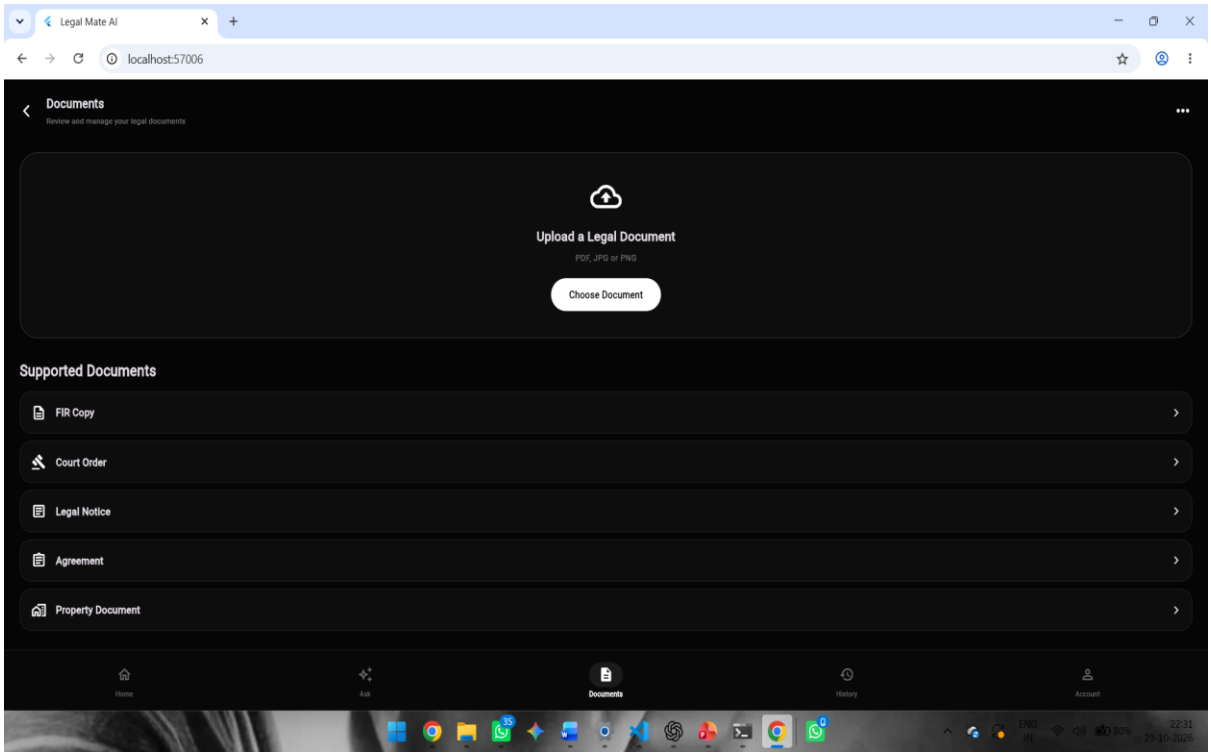



Figure 4.2 Legal document upload interface

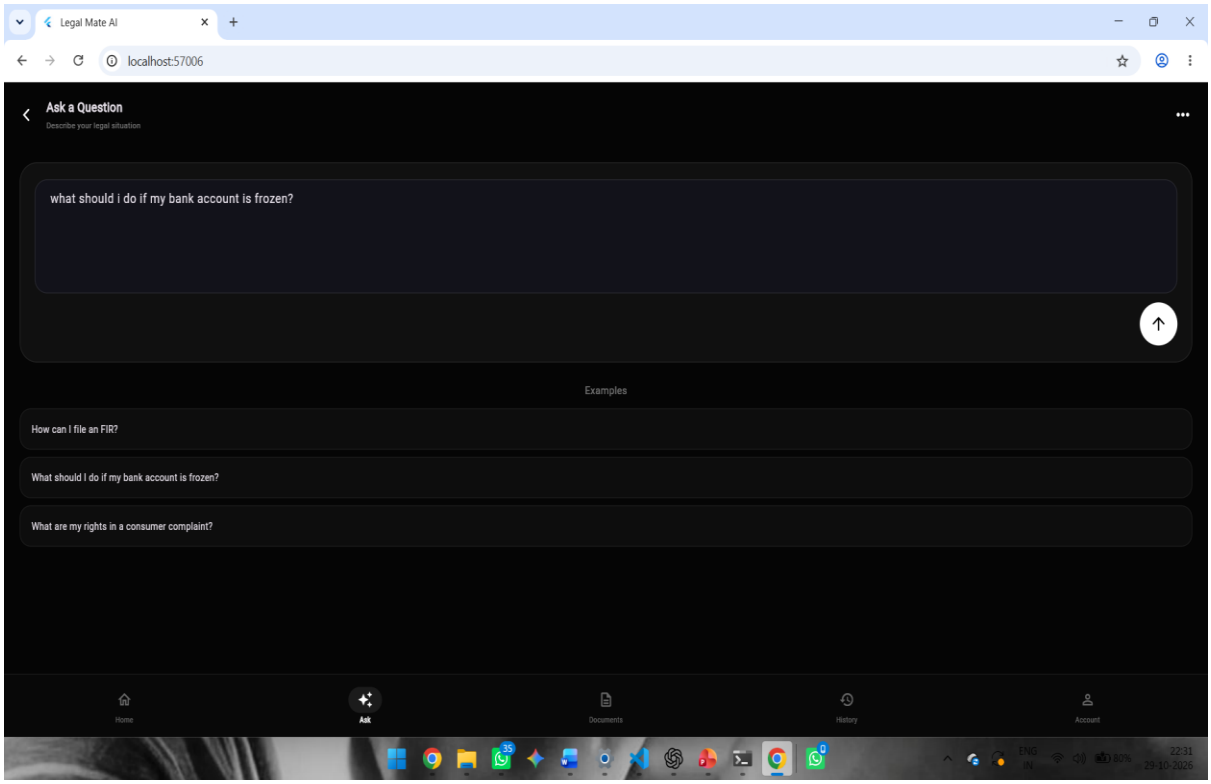



Figure 4.3 Legal query interface

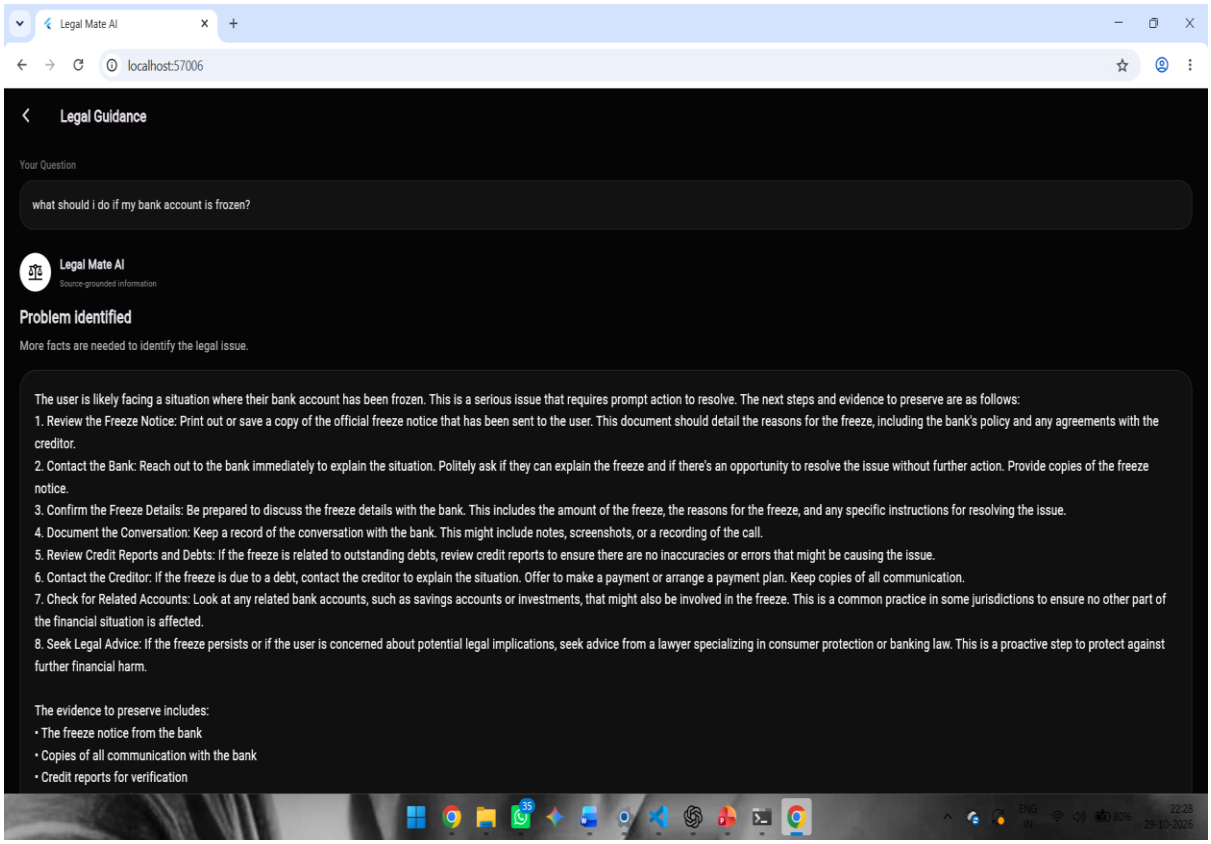



Figure 4.4 Source-grounded response interface

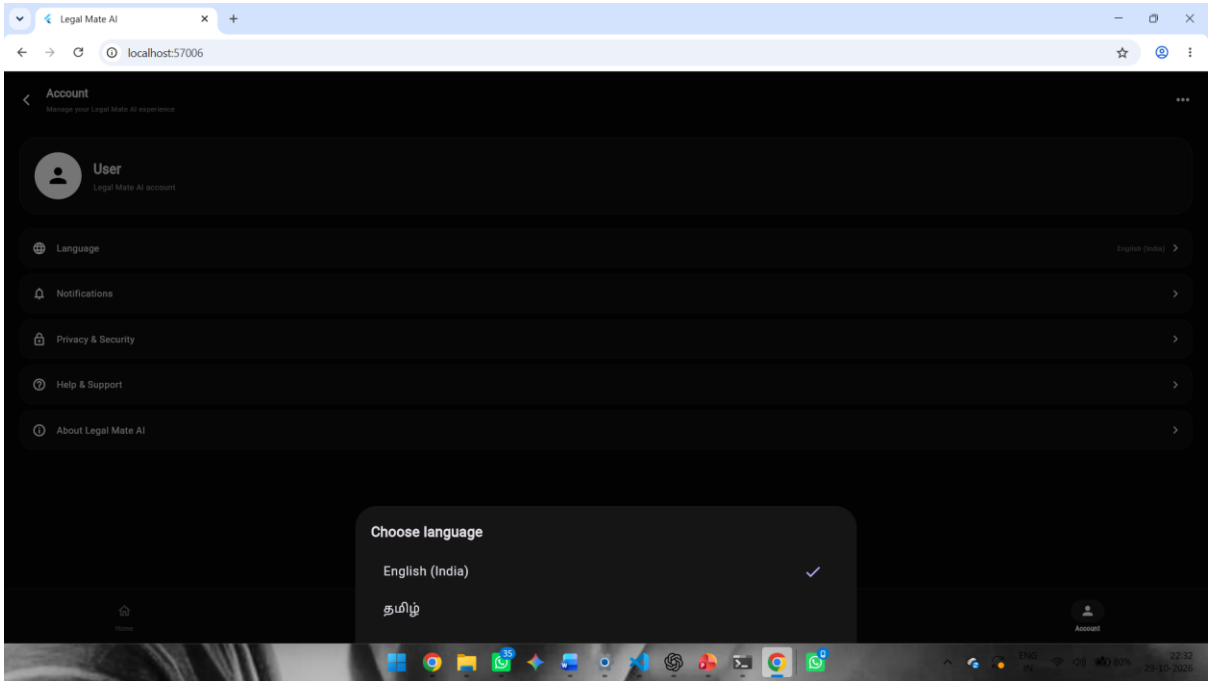



Figure 4.5 Language Selection Interface

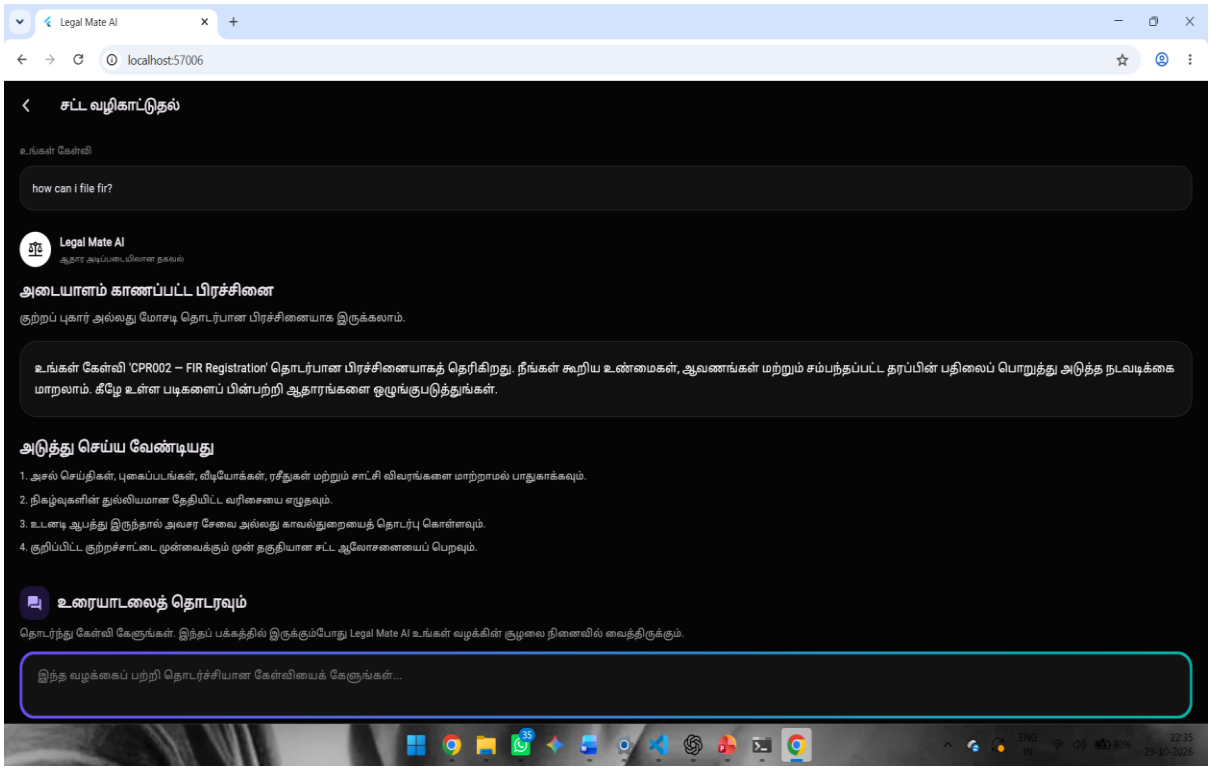



Figure 4.6 Figure 4.4 Source-grounded response interface ( Tamil )

# CHAPTER 5
# CONCLUSION AND FUTURE WORK

## 5.1 CONCLUSION

Legal Mate AI demonstrates how document processing, semantic retrieval, and generative AI can be combined to improve access to legal information. The system helps users upload legal material, retrieve relevant passages, obtain clear summaries, and ask questions connected to the content of their documents.

A central design principle is traceability. By retrieving relevant passages before generating a response, the platform can make its outputs more closely tied to the user's document. This improves usefulness for document review and legal-information discovery while reminding users to check important details.

The application is an information-support tool, not a substitute for professional legal counsel. Complex, urgent, or jurisdiction-specific legal matters should be reviewed by a qualified legal professional.

## 5.2 LIMITATIONS

- The accuracy of the system depends on the quality and availability of the legal knowledge base.
- The system may not always interpret complex or ambiguous legal queries correctly.
- The generated responses may contain incomplete or outdated legal information.
- The system supports only the languages implemented in the application, such as English and Tamil.
- Complaint drafts generated by the system require user verification before submission.
- The system provides preliminary legal information and cannot replace professional legal advice.
- The system's ability to analyse uploaded documents depends on the supported file formats and document quality.
- The system does not directly file complaints with courts or government authorities.

## 5.3 FUTURE ENHANCEMENTS

1. **Multilingual Legal-Document Assistance:** Extend the system to support additional regional languages while preserving the accuracy and meaning of legal information.
2. **Citation and Clause Comparison:** Enable document version comparison, identify modified clauses, and provide relevant legal references for better understanding.
3. **Privacy-Enhanced Deployment:** Implement stronger access controls, data encryption, document-retention settings, and audit logging to protect sensitive user information.
4. **Advanced Legal Intent Classification:** Improve the classification model to identify complex legal queries and distinguish between similar legal categories.

5. **Real-Time Legal Database Integration:** Integrate verified and regularly updated Indian legal databases to improve the accuracy and currency of retrieved legal information.
6. **Voice-Based Legal Assistance:** Introduce speech-to-text and text-to-speech features to help users interact with the system through voice commands.
7. **Improved Complaint Generation:** Enhance complaint drafting by collecting relevant incident details and generating structured drafts suitable for different complaint categories.
8. **Official Complaint Portal Integration:** Explore secure integration with authorized government grievance portals to help users access appropriate complaint submission services.
9. **Enhanced Document Processing:** Support additional document formats and improve the extraction of text from scanned documents using Optical Character Recognition (OCR).
10. **Source Verification and Reliability:** Strengthen source attribution by displaying relevant legal provisions, document references, and retrieval details alongside generated responses.
11. **Personalized Legal Guidance:** Develop a guided questionnaire that collects relevant information about a user's situation and provides more context-specific preliminary guidance.
12. **Automated Evidence Checklist:** Introduce intelligent evidence recommendations based on the type of legal issue and the details provided by the user.
13. **Performance and Accuracy Evaluation:** Conduct extensive testing using diverse legal queries to evaluate classification accuracy, retrieval relevance, response quality, and system performance.
14. **Scalability and Performance Optimization:** Optimize the backend, database, and retrieval pipeline to handle more users and larger legal-document collections efficiently.
15. **Accessibility Improvements:** Enhance the user interface with simpler navigation, readable content, and accessibility features for users with different levels of digital literacy.
16. **Legal Expert Review:** Incorporate a review mechanism through which qualified legal professionals can evaluate system responses and provide feedback to improve reliability.

These future enhancements can improve the accuracy, security, accessibility, and usability of Legal Mate AI while expanding its capabilities as an AI-powered legal assistance platform for Indian citizens.

## 5.4 FINAL REMARKS

### FINAL REMARKS

Legal Mate AI is an AI-powered legal assistance system designed to improve access to legal information for Indian citizens. By integrating Natural Language Processing, Machine Learning, and Retrieval-Augmented Generation, the system helps users understand legal issues, retrieve relevant legal information, analyse legal documents, and prepare structured complaint drafts. The support for English and Tamil aims to make legal guidance more accessible to a wider range of users. The system also provides document checklists and suggested next steps to assist users in addressing their legal concerns. Although the system has limitations in accuracy and legal interpretation, it provides a foundation for developing accessible and user-friendly legal assistance solutions. With future improvements in multilingual support, database integration, privacy, and response accuracy, Legal Mate AI has the potential to contribute to greater legal awareness and simplify access to preliminary legal guidance.

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## ANNEXURE A – CONFERENCE PAPER CERTIFICATE



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held during 21.10.2026 & 22.10.2026 organized by the Institute of CSE, SIMATS Engineering, SIMATS, Thandalam, Chennai. Your presence and engagement have contributed significantly to the discussions and knowledge-sharing on AI-driven Engineering Innovations.

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**Dr. B. Ramesh**

## ANNEXURE B – SOURCE CODE

```
import os
import json
import pandas as pd

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.svm import LinearSVC
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import (
    accuracy_score,
    precision_recall_fscore_support,
    classification_report,
)

BASE = r"C:\Users\varshan\legalmateai"
DATA = os.path.join(BASE, "data", "dataset", "final")
OUT = os.path.join(BASE, "classifier", "evaluation")

os.makedirs(OUT, exist_ok=True)

# Load datasets
train = pd.read_csv(os.path.join(DATA, "train.csv"))
validation = pd.read_csv(os.path.join(DATA, "validation.csv"))
test = pd.read_csv(os.path.join(DATA, "test.csv"))

print("Loaded datasets:")
print("Train:", len(train))
print("Validation:", len(validation))
print("Test:", len(test))

# Use only the user's query as model input
X_train = train["query"].astype(str)
X_val = validation["query"].astype(str)
```

```
X_test = test["query"].astype(str)

y_train = train["intent_id"].astype(str)
y_val = validation["intent_id"].astype(str)
y_test = test["intent_id"].astype(str)

# Encode intent IDs
encoder = LabelEncoder()

y_train_encoded = encoder.fit_transform(y_train)
y_val_encoded = encoder.transform(y_val)
y_test_encoded = encoder.transform(y_test)

# TF-IDF
vectorizer = TfidfVectorizer(
    lowercase=True,
    ngram_range=(1, 2),
    min_df=2,
    max_df=0.95,
    sublinear_tf=True,
)

X_train_tfidf = vectorizer.fit_transform(X_train)
X_val_tfidf = vectorizer.transform(X_val)
X_test_tfidf = vectorizer.transform(X_test)

print("\nTF-IDF:")
print("Training matrix:", X_train_tfidf.shape)
print("Validation matrix:", X_val_tfidf.shape)
print("Test matrix:", X_test_tfidf.shape)

# Linear SVM
model = LinearSVC(C=1.0)

print("\nTraining LinearSVC...")
```

```
model.fit(X_train_tfidf, y_train_encoded)

# Validation prediction
val_pred = model.predict(X_val_tfidf)

val_accuracy = accuracy_score(y_val_encoded, val_pred)
val_precision, val_recall, val_f1, _ = precision_recall_fscore_support(
    y_val_encoded,
    val_pred,
    average="macro",
    zero_division=0,
)

print("\n===== VALIDATION RESULTS =====")
print(f"Accuracy : {val_accuracy:.4f}")
print(f"Precision: {val_precision:.4f}")
print(f"Recall   : {val_recall:.4f}")
print(f"Macro F1 : {val_f1:.4f}")

# Final test evaluation
test_pred = model.predict(X_test_tfidf)

test_accuracy = accuracy_score(y_test_encoded, test_pred)
test_precision, test_recall, test_f1, _ = precision_recall_fscore_support(
    y_test_encoded,
    test_pred,
    average="macro",
    zero_division=0,
)

print("\n===== SEALED TEST RESULTS =====")
print(f"Accuracy : {test_accuracy:.4f}")
print(f"Precision: {test_precision:.4f}")
print(f"Recall   : {test_recall:.4f}")
print(f"Macro F1 : {test_f1:.4f}")
```

```
# Detailed classification report
report = classification_report(
    y_test_encoded,
    test_pred,
    target_names=encoder.classes_,
    zero_division=0,
)

with open(
    os.path.join(OUT, "baseline_classification_report.txt"),
    "w",
    encoding="utf-8",
) as f:
    f.write(report)

# Save metrics
metrics = {
    "model": "TF-IDF + LinearSVC",
    "train_samples": len(train),
    "validation_samples": len(validation),
    "test_samples": len(test),
    "num_intents": len(encoder.classes_),
    "validation": {
        "accuracy": val_accuracy,
        "macro_precision": val_precision,
        "macro_recall": val_recall,
        "macro_f1": val_f1,
    },
    "test": {
        "accuracy": test_accuracy,
        "macro_precision": test_precision,
        "macro_recall": test_recall,
        "macro_f1": test_f1,
    },
```

```
}

with open(
    os.path.join(OUT, "baseline_metrics.json"),
    "w",
    encoding="utf-8",
) as f:
    json.dump(metrics, f, indent=2)

print("\nSaved results to:")
print(OUT)
```